Introductory Commutative Algebra

Lecture Notes & Problem Discussions

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Abstract

Lecture notes in Commutative Algebra with references [AM69], [Bly90] and [DF04].

Contents

Chapter 1

Lecture Notes

1.1 Lecture 1

Convention. Rings are commutative with 1.

Definition 1.1. Let R, S be rings. A *morphism* φ from R to S is a map $\varphi : R \to S$ such that

(1) $\varphi(r+s) = \varphi(r) + \varphi(s)$,

(2) $\varphi(rs) = \varphi(r)\varphi(s)$,

(3) $\varphi(1) = 1$.

Convention. A subring R must contain 1.

Question. What is $R[X]$?

X is not an element of R. Consider

$$S = \{f : \mathbb{N} \to R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then S is a ring with identity

$$1_S = (1, 0, 0, \ldots, 0).$$

Define $\varphi : R \to S$ by $\varphi(a) = (a, 0, \ldots, 0)$. Then φ is a ring homomorphism, for sure.

Define $X = (0, 1, 0, \ldots)$ and observe $X^2 = (0, 0, 1, 0, \ldots)$. A polynomial P over R can then be identified as an element of S, say

$$P = (P_0, P_1, \ldots, P_n, 0, \ldots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \cdots + \varphi(P_n) \cdot X^n.$$

Definition 1.2. Let R be a ring. A non-zero, non-unit element $p \in R$ is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for $a, b \in R$, $a \mid b$ if $\exists\, r \in R$ such that $b = r \cdot a$).

Definition 1.3. Let R be a ring. A proper ideal P of R is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

Definition 1.4. Let R be a ring. $I \subseteq R$ is called an *ideal* if there exists a ring S and a morphism $\varphi : R \to S$ such that $I = \ker(\varphi)$.

Remark 1.5. This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \quad \rightsquigarrow \quad \text{the } \textit{nil-radical} \text{ of } R.$$

Theorem 1.6.

$$\mathcal{N}(R) = \bigcap_{P \in \operatorname{Spec} R} P, \qquad \text{where } \operatorname{Spec} R = \text{all prime ideals of } R.$$

Proof. Let $x \in \mathcal{N}(R)$. Then for any $P \in \operatorname{Spec} R$, there exists $n_P \in \mathbb{N}$ such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let $x \notin \mathcal{N}(R)$, and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$, as $\langle 0 \rangle \in S$, and S is a poset (ordered by inclusion). By Zorn's Lemma, let M be a maximal element of S.

Claim. M is a prime ideal.

Proof of claim. Let $a \notin M$, $b \notin M$. Now,

$$M + \langle a \rangle,\ M + \langle b \rangle \notin S \implies \exists\, m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If $ab \in M$, then $x^{m+n} \in M$, a contradiction. Hence $ab \notin M$, so M is prime. □

Since $x \notin M$ and $M \in \operatorname{Spec} R$, we conclude $x \notin \bigcap_{P \in \operatorname{Spec} R} P$, completing the proof. □

Question. What are the units of $R[X]$?

If R is an integral domain, then they are precisely the units of R.

Proposition 1.7. $f(X) = a_n X^n + \cdots + a_1 X + a_0$ is a unit if a_0 is a unit and the a_i's are nilpotent.

Proof. Let $P \in \operatorname{Spec} R$. We have a natural morphism $\varphi : R[X] \to (R/P)[X]$ given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \qquad (\overline{b_i} = b_i \bmod P).$$

Then $\varphi(f)$ is a unit in $(R/P)[X]$, and since R/P is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \qquad \text{and} \qquad \overline{a_0} \text{ is a unit of } R/P.$$

That is, $a_0 \notin P$ and $a_1, \ldots, a_n \in P$. As $P \in \operatorname{Spec} R$ was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \ \ \forall\, i \in [n], \qquad \text{for all } P \in \operatorname{Spec} R.$$

The other direction is trivial. □

1.2 Lecture 2

Definition 1.8. Let R be a ring and let $(M, +)$ be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \qquad (r, m) \longmapsto rm,$$

makes M into an *R-module* if the following hold for all $r, s \in R$ and $m, m_1, m_2 \in M$:

(i) $(r + s)(m) = rm + sm$,

(ii) $r(m_1 + m_2) = rm_1 + rm_2$,

(iii) $r(sm) = rsm = s(rm)$,

(iv) $1 \cdot m = m$.

Definition 1.9. If M, N are R-modules, a map $\varphi : M \to N$ is called an *R-module morphism* if

(i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,

(ii) $\varphi(rm) = r\varphi(m)$.

Definition 1.10. Let R be a ring and S a nonempty set. A *free R-module on S* is a pair (F, f), where F is an R-module and $f : S \to F$ is a set-theoretic map, such that: for any R-module M and any set-theoretic map $g : S \to M$, there exists a unique R-module morphism $h : F \to M$

such that

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ f = g$.

Proposition 1.11. Let (F, f) be a free R-module on S. Then:

(1) f is injective;

(2) $\mathrm{Im}(f)$ generates F as an R-module.

Proof of (1). Let $x, y \in S$ with $x \neq y$. **Goal:** $f(x) \neq f(y)$.

Take any R-module M with at least two elements, and define $g : S \to M$ such that $g(x) \neq g(y)$. By definition, there exists a (unique) morphism $h : F \to M$ such that $h \circ f = g$. Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \qquad \square$$

Definition 1.12. Let M be an R-module and $\emptyset \neq A \subseteq M$. The *submodule of M generated by A* is, equivalently:

(1) the smallest submodule of M containing A;

(2) the intersection of all submodules of M containing A;

(3) the set of all finite R-linear combinations of elements of A.

Proof of (2). Let A be the submodule of F generated by $\mathrm{Im}(f)$. Since $f(s) \in \mathrm{Im}(f) \subseteq A$ for every $s \in S$, f factors as $f = i_A \circ f^*$ for a set-theoretic map $f^* : S \to A$ and the inclusion $i_A : A \hookrightarrow F$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[dr, "f"'] & A \arrow[d, hook, "i_A"] \\ & F \end{tikzcd}$$

As (F, f) is free on S and A is an R-module, there is a unique morphism $h : F \to A$ such that $h \circ f = f^*$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[d, "f"'] & A \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

It is enough to show i_A is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So $i_A \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness in the definition of freeness (applied with $M = F$, $g = f$), $i_A \circ h = \mathrm{id}_F$. Hence i_A is onto, i.e. $A = F$, so $\mathrm{Im}(f)$ generates F. $\square$

Theorem 1.13: Uniqueness of free modules. Let (F, f) be a free R-module on a set S. Then (F', f') is also a free R-module on S if and only if there exists a unique R-module isomorphism $j : F \to F'$ such that $j \circ f = f'$.

Proof. Suppose (F', f') is also free on S. Since (F, f) is free (applied to $M = F'$, $g = f'$), there is a morphism $h : F \to F'$ with $h \circ f = f'$:

$$\begin{tikzcd} S \arrow[r, "f'"] \arrow[d, "f"'] & F' \\ F \arrow[ur, "h"'] & \end{tikzcd}$$

Likewise, since (F', f') is free (applied to $M = F$, $g = f$), there is a morphism $h' : F' \to F$ with $h' \circ f' = f$:

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "f'"'] & F \\ F' \arrow[ur, "h'"'] & \end{tikzcd}$$

Then $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$. So $h' \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness (with $M = F$, $g = f$), $h' \circ h = \mathrm{id}_F$. Similarly $h \circ h' = \mathrm{id}_{F'}$. So $j := h$ is an isomorphism with $j \circ f = f'$.

Converse. Suppose there is an isomorphism $j : F \to F'$ with $j \circ f = f'$. To show (F', f') is free on S, take any R-module M and set-theoretic map $g : S \to M$; we must produce a unique $h' : F' \to M$ with $h' \circ f' = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] & M \\ F' \arrow[ur, dashed, "h'"'] & \end{tikzcd}$$

Since (F, f) is free, there is a unique $h : F \to M$ with $h \circ f = g$. From $j \circ f = f'$ we get $j^{-1} \circ f' = f$, so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e. $h' := h \circ j^{-1}$ works:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] \arrow[dr, "f"] & M \\ F' \arrow[r, "j^{-1}"'] & F \arrow[u, "h"'] \end{tikzcd}$$

Uniqueness. Suppose $t : F' \to M$ also satisfies $t \circ f' = g$. Then $t \circ j : F \to M$ satisfies $(t \circ$

$j) \circ f = t \circ (j \circ f) = t \circ f' = g$, so by uniqueness of h (from freeness of (F, f)), $t \circ j = h$, hence $t = h \circ j^{-1} = h'$. $\square$

Theorem 1.14: Existence of free R-modules. For any ring R and nonempty set S, a free R-module on S exists.

Proof. Define

$$F = \{\theta : S \to R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For $\theta_1, \theta_2 \in F$ and $\lambda \in R$, define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \qquad (\lambda\theta)(s) = \lambda\,\theta(s).$$

One checks that F is an R-module. Define $f : S \to F$ by $f(s) = \delta_s$, where $\delta_s(t) = 1$ if $t = s$ and 0 otherwise.

Existence of h. Let M be an R-module and $g : S \to M$ a set-theoretic map. Define $h : F \to M$ by

$$h(\theta) = \sum_{s \in S} \theta(s)\, g(s),$$

a finite sum for every $\theta \in F$. For any $t \in S$,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t)\, g(t) = g(t),$$

so $h \circ f = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Uniqueness of h. For $\theta \in F$ and $t \in S$,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s)\, \delta_s(t) = \left(\sum_{s \in S} \theta(s)\, \delta_s\right)(t) \qquad \forall\, t \in S,$$

so $\theta = \sum_{s \in S} \theta(s)\, \delta_s$. Now let $h' : F \to M$ be any R-module morphism with $h' \circ f = g$. Then for $\theta \in F$,

$$h'(\theta) = h'\left(\sum_{s \in S} \theta(s)\, \delta_s\right) = \sum_{s \in S} \theta(s)\, h'(\delta_s) = \sum_{s \in S} \theta(s)\, h'(f(s)) = \sum_{s \in S} \theta(s)\, g(s) = h(\theta).$$

Hence $h = h'$. $\square$

Chapter 2

Problem Discussions

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